Cyclic kernel of a K3-type action bounded by canonical Fano Weil index (Proposition 1.9)

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Editorial scope. One author proposition and full current supporting proof chain; no mathematical repair, independent certification or formal admission.

Allowed standard prerequisites (editorial).

- Faithful tangent action at a fixed point. Popov [Pop14], Lemma 4; author Lemma 2.6 (lines 491–494). Standard input to the fully reproduced fixed-divisor and cyclic-kernel proofs.
- MMP singularities, inversion of adjunction, adjunction, vanishing and Serre duality. Kollar–Mori [KM98], especially Corollary 5.25 explicitly cited at line 2730; author lines 2725–2740. External tools, while the actual quotient construction, torsion contradiction and canonical discrepancy calculation are retained.
- Cyclic branched-cover divisibility. J. J. Wavrik, Deformations of branched coverings of complex manifolds, Amer. J. Math. 90 (1968), 926–960, Theorem 1.2; DOI 10.2307/2373491; author citation [Wav68] at line 2742. Applied by the author to the smooth locus Y^0, giving m B^0 linearly equivalent to D^0. The passage Pic(Y^0)=Cl(Y^0)=Cl(Y) and closure B is reproduced, not supplied editorially.
- Bounded canonical Fano index. Birkar [Bir21], DOI 10.4007/annals.2021.193.2.1; Jiang–Liu [JL25], arXiv:2508.16364v1; Suzuki [Suz04], DOI 10.1007/s00229-004-0442-4. Existence of the maximal index is standard external boundedness input; Remark 11.4 retains the author's bounds 19 (terminal case) and 66. No effective-index theorem is separately counted.

Source and scope. Konstantin Loginov, *Finite abelian groups acting on rationally connected threefolds I: Groups of product type*, arXiv:2408.11645v3 (22 August 2026), DOI 10.46298/epiga.2026.15051. Exact author-native source under CC BY-SA 4.0; this extracted adaptation is distributed under the same license with attribution. Changes are selection, assembly, standard formatting and restored source counters; mathematical wording is unchanged.

Only Proposition 1.9 is the counting target. All retained preliminary results, Lemma 11.2, Proposition 11.3 and Remark 11.4 are support, not additional items. The complete current quotient and torsion-elimination arguments are reproduced. Source numbering is retained across omitted passages; the full original bibliography is included to retain author citation labels and repeated-author conventions.

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1 Introduction

We work over the field of complex numbers $\mathbb{C}$.

Finite abelian groups are among the most elementary objects studied in algebra. In algebraic geometry, rational varieties form a rather simple and well-behaved class of varieties. Consider the following question.

Question 1.1. Which finite abelian groups can faithfully and biregularly act on rational (or rationally connected) varieties of a given dimension?

Since we consider varieties endowed with the action of a finite group, there exists an equivariant resolution of singularities, so we can consider the class of finite abelian groups that act on rational (or rationally connected) smooth projective varieties. Also, the notions of biregular and birational group actions would give us the same list of groups in answer to Question 1.1 since we can resolve the indeterminacy of the group action.

In this paper, by a K3 surface we mean a normal projective surface S with at worst canonical (that is, du Val) singularities such that $\mathrm{H}^1(S, \mathscr{O}_S) = 0$ and $K_S \sim 0$. We say that a finite abelian group G is of *K3 type* if G faithfully acts on a $G\mathbb{Q}$-Fano threefold X preserving a K3 surface $S \in |-K_X|$ with at worst du Val singularities. In particular, we have that G is an extension of a finite abelian group H that faithfully acts on a K3 surface by a cyclic group (cf. Corollary 2.9):

$$0 \longrightarrow \mathbb{Z}/m \longrightarrow G \longrightarrow H \longrightarrow 0.$$

Note that if a finite group faithfully acts on a singular K3 surface, then this action lifts to its minimal resolution. Our main result is the following theorem (see Section 2.4 for the definitions of a $G\mathbb{Q}$-Fano variety and a $G\mathbb{Q}$-Mori fiber space).

Our next result is a boundedness property for groups of K3 type. By the Weil index of a Fano variety X with at worst canonical singularities, we mean the number $C(X) = \max\{k \mid -K_X \sim kH\}$, where H is a Weil divisor on X.

Proposition 1.9. *Assume that G is a group of K3 type. Then in the corresponding exact sequence*

$$0 \longrightarrow \mathbb{Z}/m \longrightarrow G \longrightarrow H \longrightarrow 0,$$

where the group H faithfully acts on a K3 surface, we have $m \leq C$, where C is the maximal Weil index of a Fano threefold with at worst canonical singularities.

All the varieties are projective and defined over the field of complex numbers $\mathbb{C}$, unless stated otherwise. We use the language of the minimal model program (the MMP for short); see *e.g.* [KM98].

2.1 Contractions

By a *contraction* we mean a projective morphism $f\colon X \to Y$ of normal varieties such that $f_*\mathscr{O}_X = \mathscr{O}_Y$. In particular, f is surjective and has connected fibers. A *fibration* is defined as a contraction $f\colon X \to Y$ such that $\dim Y < \dim X$.

Let G be a finite group. By a *G-variety* we mean a variety X together with an action of a group G. If f is a G-equivariant contraction (resp. a G-equivariant fibration) of G-varieties, we call it a *G-contraction* (resp. a *G-fibration*).

2.2 Pairs and singularities

A *pair* (X, B) consists of a normal variety X and a boundary Weil $\mathbb{Q}$-divisor B with coefficients in $[0, 1]$ such that $K_X + B$ is $\mathbb{Q}$-Cartier. Let $\phi\colon W \to X$ be a log resolution of (X, B), and let

$$K_W + B_W = \phi^*(K_X + B)$$

be the log pullback of (X, B). The *log discrepancy* of a prime divisor D on W with respect to (X, B) is $1 - \operatorname{coeff}_D B_W$, and it is denoted by $a(D, X, B)$. We say (X, B) is lc (resp. klt, ϵ-lc) if $a(D, X, B) \geq 0$ (resp. > 0, $\geq \epsilon$) for every D. Note that if (X, B) is ϵ-lc, then automatically $\epsilon \leq 1$ because $a(D, X, B) = 1$ for almost all D. We say that the pair (X, B) is *strictly lc* if it is lc but not klt. We say that the pair (X, B) is *plt* if $a(D, X, B) > 0$ holds for any ϕ-exceptional divisor D and for any log resolution ϕ. We say that the pair (X, B) is *dlt* if $a(D, X, B) > 0$ holds for any ϕ-exceptional divisor D and for some log resolution ϕ.

An *lc-place* of (X, B) is a prime divisor D over X, that is, on a birational model of X, such that $a(D, X, B) = 0$. An *lc-center* is the image on X of an lc-place.

2.3 Log Fano and log Calabi–Yau pairs

Let (X, B) be an lc pair. We say (X, B) is a *log Fano pair* if $-(K_X + B)$ is ample. If $B = 0$, then X is called a *Fano variety*. We say that an lc pair (X, B) is a *log Calabi–Yau pair* if $K_X + B \sim_{\mathbb{Q}} 0$. If $B = 0$, then X is called a *Calabi–Yau variety*.

2.4 Mori fiber space

Let G be a finite group. Recall that a G-variety X is called $G\mathbb{Q}$-factorial if every G-invariant Weil divisor on X is $\mathbb{Q}$-Cartier. A *$G\mathbb{Q}$-Mori fiber space* is a $G\mathbb{Q}$-factorial variety X (we assume that the action of G on X is faithful) with at worst terminal singularities, together with a G-contraction $f\colon X \to Z$ to a variety Z such that $\rho^G(X/Z) = 1$ and $-K_X$ is ample over Z. If X is G-factorial (*e.g.* if X is smooth), we call it a *G-Mori fiber space*. If Z is a point, we say that X is a $G\mathbb{Q}$-Fano variety. If $\dim X = 2$ and $\dim Z = 1$, then we say that $f\colon X \to Z$ is a *G-conic bundle*. If $\dim X = 2$ and Z is a point, then we say that $f\colon X \to Z$ is a *G-del Pezzo surface*.

2.7 Group actions on algebraic varieties.

We collect some generalities on the actions of finite groups on algebraic varieties.

Definition 2.4. Let G be a finite group. By $\mathfrak{r}(G)$, we denote the *rank* of G, that is, the minimal number of generators of G.

Remark 2.5. Note that if G is a finite abelian group and $H \subset G$ is a subgroup, then $\mathfrak{r}(H) \leq \mathfrak{r}(G)$.

Lemma 2.6 (*cf.* [Pop14, Lemma 4])**.** *Let X be an algebraic variety, and let G be a finite group such that $G \subset \mathrm{Aut}(X)$. Assume $P \in X$ is a fixed point of G. Then the induced action of G on the tangent space T_PX is faithful.*

Corollary 2.7. *If a finite abelian group $G \subset \mathrm{Aut}(X)$ has a fixed point $P \in X$, where X is an algebraic variety of dimension n and P is a smooth point, then $\mathfrak{r}(G) \leq n$.*

In what follows, we will need the following results.

Lemma 2.8 (*cf.* [Pro11, Lemma 2.6])**.** *Let X be a 3-dimensional algebraic variety X with isolated singularities, and let G be a finite abelian group such that $G \subset \mathrm{Aut}(X)$.*

(1) *If there is a curve $C \subset X$ of G-fixed points, then $\mathfrak{r}(G) \leq 2$.*

(2) *If there is a (possibly, reducible) divisor $S \subset X$ of G-fixed points, then $\mathfrak{r}(G) \leq 1$. If moreover S is singular along a curve, then G is trivial.*

Proof. We prove the first claim. Let P be a general point in C. Since X has isolated singularities, P is smooth on X. In particular, $\dim T_PX = 3$. By Lemma 2.6, the induced action of G on T_PX is faithful. On the other hand, the action of G on the 1-dimensional subspace $T_PC \subset T_PX$ is trivial, since by assumption the action of G is trivial on C. Hence G faithfully acts on a G-invariant complement T' to T_PC in T_PX. So we have $\dim T' = 2$. Since G is a finite abelian group, it follows that $\mathfrak{r}(G) \leq 2$ in this case.

The first part of the second claim is proven analogously. We prove the second part. By assumption, the action of G on S it trivial. It follows that the induced action of G on T_PS is trivial for any point $P \in S$. For a general point $P \in \mathrm{Sing}(S)$, we have $T_PS = T_PX$ since X is smooth at P and P is singular on S. By Lemma 2.6, we conclude that G is trivial. □

Corollary 2.9. *If a finite abelian group G faithfully acts on a 3-dimensional algebraic variety X with isolated singularities preserving a surface S, then there is an exact sequence*

$$0 \longrightarrow \mathbb{Z}/m \longrightarrow G \longrightarrow H \longrightarrow 0$$

for $m \geq 1$, where $H \subset \mathrm{Aut}(S)$.

Proof. Denote the image of a natural map $\phi\colon G \to \mathrm{Aut}(S)$ by H. By Lemma 2.8(2), the kernel of ϕ is cyclic. The claim follows. □

11 Groups of K3 type

In this section, we work with finite abelian groups of K3 type. Recall that by a K3 surface we mean a normal projective surface S with at worst canonical (that is, 1-lc) singularities such that $\mathrm{H}^1(S, \mathscr{O}_S) = 0$ and $K_S \sim 0$.

Definition 11.1. We say that a finite abelian subgroup G is of *K3 type* if G faithfully acts on a $G\mathbb{Q}$-Fano threefold X preserving a K3 surface $S \in |-K_X|$ with at worst du Val singularities.

Let X be a $G\mathbb{Q}$-Fano threefold, where G is a finite abelian group and G acts on X faithfully. Assume that there exists a G-invariant element $S \in |-K_X|$ that is a K3 surface with at worst canonical singularities. Consider the exact sequence

$$0 \longrightarrow K \longrightarrow G \longrightarrow H \longrightarrow 0,$$

where $H \subset \mathrm{Aut}(S)$ and K fixes S pointwise. Note that $K = \mathbb{Z}/m$ for some $m \geq 1$ (*cf.* Corollary 2.9). In particular, the group G is of K3 type. Let $Y = X/K$ be the quotient, and let $f\colon X \to Y$ be the natural projection.

Lemma 11.2. *Assume that K is non-trivial. Then the action of K is free in codimension 1 outside S. Hence, the divisorial part of the ramification of f is precisely S.*

Proof. Let K' be a non-trivial subgroup of K. Assume there exists a K'-fixed divisor D such that $D \neq S$. Then K' fixes $S \cup D$ pointwise. Since S is ample, we have $S \cap D \neq \varnothing$. Since S is $\mathbb{Q}$-Cartier, there exists a curve $C \subset S \cap D$. At a general point of C, the variety X is smooth, so the tangent space to $S \cup D$ at some point $P \in C$ is equal to the tangent space to X. However, this contradicts Lemma 2.6 applied to the group K'. □

Proposition 11.3. *The threefold $Y = X/K$ is an $H\mathbb{Q}$-factorial Fano variety with at worst canonical singularities. Moreover, $-K_Y \sim mB$ for some $B \in \mathrm{Cl}(Y)$.*

Proof. Recall that we have $K = \mathbb{Z}/m$ for some $m \geq 1$. We may assume that $m > 1$. First we show that Y is $H\mathbb{Q}$-factorial. Indeed, this follows from the fact that

$$\mathrm{Cl}(Y) \otimes \mathbb{Q} = (\mathrm{Cl}(X) \otimes \mathbb{Q})^K$$

and hence

$$(\mathrm{Cl}(Y) \otimes \mathbb{Q})^H = (\mathrm{Cl}(X) \otimes \mathbb{Q})^G = (\mathrm{Pic}(X) \otimes \mathbb{Q})^G.$$

Put $D = f(S)$. Since K fixes S pointwise, we have $S \simeq D$. The Hurwitz formula yields

$$0 \sim K_X + S = f^*(K_Y + D),$$

and hence $m(K_Y + D) \sim 0$. Note that Y is a Fano variety with klt singularities. By inversion of adjunction, the pair (Y, D) is plt. Observe that the divisor D is $\mathbb{Q}$-Cartier. Put $T = K_Y + D$. We show that $T \sim 0$. Assume otherwise. Since $mT \sim 0$, the divisor T is non-trivial torsion, and hence $\mathrm{H}^0(Y, \mathscr{O}_Y(T)) = 0$. By adjunction (*cf.* [KM98, Corollary 5.25]), there is an exact sequence

$$0 \longrightarrow \mathscr{O}_Y(K_Y) \longrightarrow \mathscr{O}_Y(T) \longrightarrow \mathscr{O}_D(K_D) \longrightarrow 0.$$

Kawamata–Viehweg vanishing and Serre duality give $\mathrm{H}^1(Y, \mathscr{O}_Y(K_Y)) = 0$. It follows that $\mathrm{H}^0(D, \mathscr{O}_D(K_D)) = 0$, which contradicts $K_D \sim 0$. Thus $K_Y + D \sim 0$.

We have shown that $-K_Y \sim D$. Since $K_Y + D$ is Cartier and (Y, D) is plt, the log discrepancy $a(E, Y, D)$ of every exceptional divisor E over Y is a positive integer. Therefore,

$$a(E, Y) = a(E, Y, D) + \mathrm{ord}_E(D) \geq 1,$$

so Y has canonical singularities.

We claim that there exists a Weil divisor B such that $mB \sim D$. Indeed, let Y^0 be a smooth locus of Y, and let X^0 be its preimage via the map f. We denote the corresponding map by $f^0\colon X^0 \to Y^0$. Put $D^0 = D \cap Y^0$. Then applying [Wav68, Theorem 1.2], we see that there exists an element $B^0 \in \mathrm{Pic}(Y^0)$ such that $mB^0 \sim D^0$. Since $\mathrm{Pic}(Y^0) = \mathrm{Cl}(Y^0) = \mathrm{Cl}(Y)$, we see that $mB \sim D$, where B is the closure of B^0 in Y. □

Remark 11.4. In the above notation, if Y is terminal (this holds for example if the action of K is free outside S), then by [Suz04] we have $m \leq 19$. In general, by [JL25] we have $m \leq 66$.

Proof of Proposition 1.9. This follows from Lemma 11.2, Proposition 11.3, and Remark 11.4. □

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